9.2 Ramsey's Theorem in Miniature

WORKSHEET

The question

Six points A, B, C, D, E and F lie in three dimensional space and are in general positions, that is, no three are collinear and no four lie on a plane. All possible line segments joining pairs of points are drawn and coloured either gold or silver. Prove that there is a triangle whose edges are entirely of one colour. [**Hint**: consider segments radiating from A.]

Give a sketch showing that the result is false for five points in general positions.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself The six points matter only through which pairs are joined by which colour: the geometry is a distraction. How many segments are there, and how many leave each point?

STAGE 1 Five segments, two colours

Ask yourself Five segments leave A, each gold or silver. What can you say for certain about their colours?

STAGE 2 Look at the far ends

Ask yourself Now look at the triangle BCD. What happens if any one of its sides is gold? What if none of them is?

STAGE 3 Five points are not enough

Ask yourself With five points, each point has only four segments. How might you colour them so that each point has exactly two of each colour?

Try this next

1. **At least two.** Show that in any two-colouring of K_6 there are in fact at least *two* one-colour triangles. (*Hint:* count the "bichromatic angles": pairs of edges of different colours at a common vertex. A point with g gold and $5-g$ silver edges is at the apex of $g(5-g) \le 6$ such angles. Each triangle that is not one-colour contains exactly two.)

2. **Three colours.** If the edges of K_{17} are coloured in three colours, show that there is a one-colour triangle. (*Hint:* one point has at least six edges of one colour.)

3. **Three or four.** Show that $R(3,4) \le 10$: in any gold/silver colouring of K_{10} there is a gold triangle or a silver K_4. (*Hint:* a point has either at least 4 gold edges or at least 6 silver edges.) In fact $R(3,4) = 9$.

4. **Monotone subsequences.** Prove the Erdős–Szekeres theorem for $n = 2$: any sequence of 5 distinct numbers contains an increasing or decreasing subsequence of length 3. Find a sequence of 4 with neither.

Further reading

- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific), the chapters on the pigeonhole principle and Ramsey theory.
- **Ronald Graham & Joel Spencer, "Ramsey theory"**, *Scientific American* (July 1990). A popular introduction by two leading researchers.
- **Noga Alon & Joel Spencer, *The Probabilistic Method*** (Wiley). The standard book on Erdős's method; the first chapter is accessible.
- **Paul Hoffman, *The Man Who Loved Only Numbers*** (Fourth Estate). A biography of Erdős.

The full guided solution, and what the question leads to at university, is in Section 9.2:
https://beyondstep.keithmcnulty.org/ch09/ramsey.html